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Hybrid CA-SAE-AFB and ensemble learning framework for accurate heart disease prediction

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Abstract

Heart disease prediction from structured clinical records requires models that can represent nonlinear feature interactions while avoiding optimistic estimates caused by data leakage and model-selection bias. This study presents a hybrid CA-SAE-AFB framework that combines a sparse autoencoder, adaptive latent-feature re-weighting, logistic regression, XGBoost, out-of-fold stacking, validation-only F1 threshold optimization, and a two-stage decision rule. The study includes a 50-seed stratified analysis for stability assessment and uses leakage-controlled nested stratified 5-fold outer cross-validation with a 3-fold inner loop as the primary evaluation protocol. All preprocessing, representation learning, threshold selection, and meta-learner fitting are restricted to the appropriate training/inner-validation partitions, while outer test folds remain untouched until final evaluation. Under the nested evaluation, CA-SAE-AFB achieved a mean accuracy of 0.8617 (95% CI: 0.8365–0.8869), F1-score of 0.8772 (95% CI: 0.8591–0.8952), MCC of 0.7208 (95% CI: 0.6695–0.7721), and ROC-AUC of 0.9153 (95% CI: 0.8944–0.9361). The repeated-split and nested-CV estimates were closely aligned. Component ablation showed the clearest directional contribution from adaptive feature boosting, whereas the effects of the sparse autoencoder and meta-learner were metric-dependent. On an NVIDIA Tesla T4 GPU, the complete single-sample inference pipeline required 83.18 ms on average over 200 timed runs after 50 warm-up runs. These results support computational feasibility and methodological robustness for further investigation in clinical decision-support settings, while external prospective clinical validation remains necessary.

Keywords Deep learning · Heart disease prediction · Machine learning · Sparse autoencoder

Abbreviations

AFB	Adaptive Feature Boosting
AI	Artificial Intelligence
CA-SAE-AFB	Clinical Adaptive Sparse Autoencoder with Adaptive Feature Boosting
CNN	Convolutional Neural Network
CV	Cross-Validation
CVD	Cardiovascular Disease

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DL	Deep Learning
GB	Gradient Boosting
GPU	Graphics Processing Unit
LR	Logistic Regression
MCC	Matthews Correlation Coefficient
MI	Mutual Information
ML	Machine Learning
OOF	Out-of-Fold
PR-AUC	Precision–Recall Area Under the Curve
ROC-AUC	Receiver Operating Characteristic–Area Under the Curve
SAE	Sparse Autoencoder
XGBoost	Extreme Gradient Boosting

Introduction

Background and motivation

Cardiovascular disease (CVD) is a common term used for different types of diseases related to the heart and blood vessels [1]. It refers to a condition that affects how the heart pumps blood or damages the blood flow system in the human body [2]. Under this main group, CVD includes several specific problems like coronary artery disease, stroke, and rheumatic heart disease. Most of the time, these diseases share a main cause, which is the building up of fat and blockages in the blood vessels, known as atherosclerosis [3]. Machine learning and deep learning algorithms help medical professionals diagnose heart diseases. These techniques can process complex clinical information. The conventional machine learning methods, including logistic regression, support vector machines, and decision trees, formed a core part of earlier studies. These are simple and interpretable but have poor ability to associate complicated variables in medical data [4, 5]. As a way to address these limitations, many have studied new approaches based on deep learning models, which are capable of automatically learning high-level features from raw data and improving prediction accuracy [6].

Deep neural networks and hybrid architectures have shown potential in heart disease prediction tasks. Hybrid models that integrate various learning techniques, such as recurrent neural networks and convolutional neural networks, have shown better results because they can learn spatial and temporal dependencies [7]. Moreover, feature engineering and feature augmentation methods have also been presented to improve the quality of input information and classification results [2, 8]. Learning meaningful representations of medical data has also been popularly performed with the help of autoencoders. Such techniques can reduce dimensions and isolate significant features, to provide more robust predictions [9, 10]. Moreover, the latest research has included explainable artificial intelligence and privacy-conserving methods to ensure the models are more trustworthy and applicable in real-life healthcare contexts [11, 12]. Despite these developments, more powerful and effective models that can accommodate the interaction of complex features, enhance generalization, and give consistent performances with various datasets are still required. These limitations motivate the design of ensemble-based and hybrid techniques for heart disease prediction.

Problem statement

There are still several issues that need to be addressed, despite the development of many machine learning and deep learning models for predicting cardiac diseases. Overfitting of deep learning models also poses a challenge. Complex neural network structures tend to perform well on training data but not necessarily on unseen data.

This can be due to limited data and inadequate validation methods. The problem of feature representation is also important. Although feature engineering and autoencoder-based approaches have been employed, most of the approaches fail to comprehensively model the interactions between various clinical variables. This restricts the capability of the model to comprehend complicated connections in the information.

Moreover, the current approaches are based on predetermined classification limits, which might not be the best with medical decision-making. Lack of threshold optimization may influence the sensitivity/specificity compromise, which is significant in medical practice. Also, there are numerous studies that concentrate on single models instead of using a combination of multiple models in a systematic manner. Despite the hybrid methods that have been studied, work has not been done on effective ensemble strategies that will be used to combine various learning paradigms to enhance the overall performance [13, 14]. These obstacles outline the necessity of a more powerful and effective prediction framework that will overcome the drawbacks of the current techniques.

Despite significant advancements in machine learning and deep learning techniques for heart disease prediction, several limitations persist. Existing models often rely on either handcrafted features or deep representations without effectively integrating both. Additionally, most approaches employ fixed decision thresholds, which are suboptimal for medical diagnosis due to the imbalance between false positives and false negatives. Furthermore, limited attention has been given to enhancing feature importance within latent representations and ensuring prediction stability across multiple data splits. These challenges highlight the need for a robust framework that integrates feature learning, adaptive feature boosting, ensemble modeling, and threshold optimization.

Research objectives and contributions

This work aims to develop a robust and reliable model for heart disease prediction using advanced machine learning and deep learning methods. The contributions are as follows:

- A hybrid CA-SAE-AFB model is proposed to learn better feature representations using a sparse autoencoder with adaptive feature boosting.
- A stacking ensemble is used by combining logistic regression and XGBoost, with a gradient boosting meta-learner to improve prediction.
- An F1-score based threshold and a two-stage decision method are used to make predictions more reliable.
- The model is tested using multiple iterations to ensure stable and consistent performance.

The novelty of the proposed work lies in the integration of adaptive feature boosting within a sparse autoencoder framework, enabling dynamic weighting of latent features based on their relevance. Unlike traditional autoencoders, the proposed CA-SAE-AFB model incorporates an attention-like mechanism to enhance discriminative feature representation. Furthermore, the combination of latent-space learning and original feature modeling through a stacking ensemble improves both linear and nonlinear pattern capture. The inclusion of F1-score-based threshold optimization and a two-stage decision strategy further enhances classification reliability, particularly in borderline cases.

Figure 1 provides an overview of the proposed framework for heart disease prediction. The system begins with raw clinical data, which undergoes preprocessing, including categorical encoding, feature transformation, and standardization. Data-quality checks are also performed to identify missing observations. Subsequently, a feature interaction layer is introduced to capture relationships among clinical variables and enhance the quality of input data representation. The processed data is converted into latent feature representations by the proposed CA-SAE-AFB mechanism, which is effective for extracting meaningful patterns and dimensionality reduction. After that, the learned features are used in the model-building stage, where several learning strategies are combined to enhance predictive performance. The final output is a probability score generated during the model-building stage. A threshold optimization mechanism is also used to determine the optimal decision boundary to further optimize the performance of classification. Predictions are refined in a two-stage decision method as part of the decision approach, followed by a probability estimation stage. The final stage produces the prediction outcome.

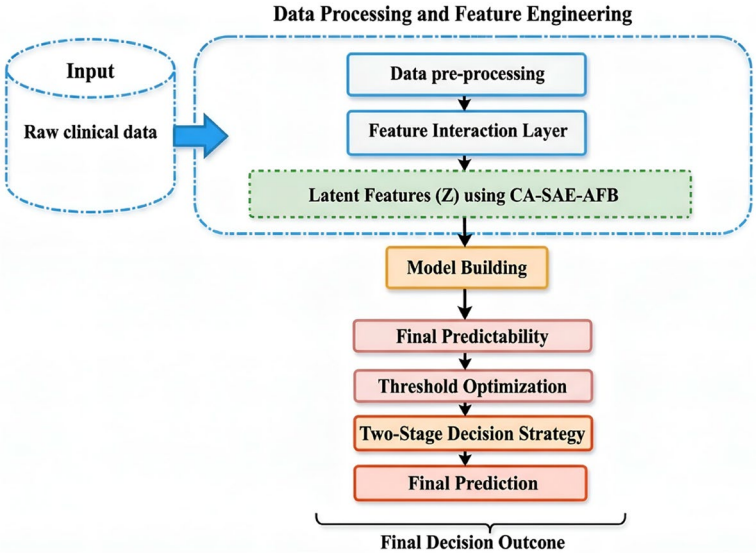


Fig. 1 Overview of the proposed framework for heart disease prediction

This architecture emphasizes the main components of the proposed work, such as feature interaction, hybrid feature learning, ensemble-based prediction, and threshold optimization, which are combined into one framework for heart disease prediction.

Organization of the paper

The remaining part of the paper is structured as follows: Related work section gives a detailed account of the literature review of the heart disease prediction domain, with the conventional machine learning, deep learning, stacking, and ensembling as well as limitations of each. System model / architecture section explains the general system architecture of the proposed CA-SAE-AFB framework, including its key components and design principles. Problem formulation section formulates the problem and defines the major assumptions, constraints, and optimization aims used in the model. Proposed methodology section describes the proposed methodology, which involves data preprocessing, feature interaction, a sparse autoencoder with adaptive feature boosting, an ensemble strategy, and threshold optimization mechanisms. Implementation details section describes the implementation details, including the experimental setup, dataset, parameter settings, and tools used for model development. Performance evaluation and results section contains the performance assessment and discussion of the experimental findings based on several evaluation criteria and a comparative analysis with the baseline models. Discussion section discusses the main observations, ablation findings, statistical analysis, and limitations of the study. Conclusion and future work section concludes the paper and gives possible directions for future research.

Related work

Existing approaches and techniques

Both traditional machine learning and modern deep learning methods have been extensively used to predict heart diseases. Neural networks, including multilayer perceptrons, were employed to model nonlinear connections within clinical information and assist in making medical decisions [15]. These methods proved the possibilities of artificial intelligence in healthcare only because of deep architectures and feature representation capabilities.

In recent years, there has been development of machine learning methods, including logistic regression, support vector machines, and random forests, which are used in heart disease datasets. These techniques are simple,

interpretable, and computationally efficient. Studies have demonstrated that prediction can be enhanced by using several machine learning algorithms and, in particular, appropriate feature selection methods to improve accuracy [4, 5]. Feature engineering has also been identified as a key step in which input features are carefully designed to improve model performance [8]. Feature learning is one of the most important aspects of deep learning models that attracts a lot of attention. Fully connected deep neural networks have been utilized in clinical decision-support systems for heart disease prediction [6]. These models can automatically learn high-level representations. However, they can be difficult to tune and may be prone to overfitting, particularly with limited data.

More complex architectures, such as convolutional neural networks and long short-term memory networks, have also been investigated. Research on fusion models based on CNN has also been done, and LSTMs have been found to perform better because they reproduce the spatial and temporal patterns of medical data [7]. Additionally, explainable deep learning models have been incorporated with techniques of artificial intelligence to accomplish and enhance clinical application transparency and trust [11]. In addition, deep learning architectures have been extended using advanced network designs [16]. Convolutional neural networks for automatic feature extraction have also been proposed a hybrid deep dense architecture to enhance predictive capability. The prediction performance of hybrid deep learning frameworks has also been enhanced by integrating several neural network structures. Strong frameworks of hybrid deep neural networks have been suggested to improve the learning and classification of features and accuracy [14]. Similarly, deep learning models of feature augmentation methods have been applied in order to achieve more input data and improved prediction outcomes [2].

Autoencoders have been extensively applied in feature learning and dimensionality reduction. Sparse autoencoders are able to encode meaningful latent representations based on clinical data, which enhances classification [9]. Recent works have incorporated autoencoders with advanced architectures such as DenseNet to improve feature extraction [17]. Moreover, autoencoder-based feature engineering methods can provide useful representations for cardiovascular disease prediction [10].

Improving performance, security, and privacy has also been a topic of recent research. Differential privacy has been offered to defend sensitive privacy in medical data and preserve model performance [12]. These major concerns need to be dealt with the real-life healthcare applications where data privacy is required. Ensemble learning is also a possibility and has been researched to increase the accuracy of prediction. Hybrid ensemble approaches that involve simultaneous usage of a number of models and feature selection algorithms like soft voting and hard voting on a collection of machine learning methods have been encouraging in forecasting heart disease propensity [13]. These approaches take advantage of the strengths of various models to improve reliability.

Recent cardiovascular prediction studies have also explored optimizer-guided feature selection and ensemble learning. Kumar and Rekha developed a dense-network framework with a Gaussian-optimizer-based feature-selection strategy for cardiovascular disease prediction [18]. Their work illustrates the potential of meta-heuristic feature optimization, while the present study instead emphasizes leakage-controlled validation and adaptive latent-feature re-weighting. Kumar and Rekha subsequently proposed an improved Hawks-Optimizer-based cardiovascular prediction framework using a stacked learning strategy [19]. Arunachalam and Rekha investigated ensemble machine-learning models for cardiovascular disease prediction and reported the value of combining complementary classifiers and feature-selection mechanisms [20]. More recently, Saranya et al. proposed DenseNet-ABiLSTM for multiclass arrhythmia detection using PPG signals, demonstrating the growing role of hybrid deep-learning architectures in cardiovascular intelligence [21]. These studies provide relevant context for optimizer-based feature selection, ensemble learning, and hybrid deep architectures, while CA-SAE-AFB focuses on structured tabular heart-disease prediction with fold-local preprocessing, OOF stacking, and nested evaluation.

Comparative analysis of machine learning and deep learning models

The relative merits of traditional machine learning and deep learning models remain an important consideration in heart-disease prediction. Traditional models, such as logistic regression, naive Bayes, decision trees, support

vector machines, and random forests, remain useful baselines because of their transparency, relatively low computational cost, and effectiveness on small to moderate-sized tabular clinical datasets [4, 22].

Recent works have also tackled the feature selection and early-stage prediction-based methods. The authors [23] suggested a feature-selection-based model for early detection of heart disease, and Srinivasan et al. [24] suggested an active learning model to optimize classification performance with limited labeled data. These models have been especially popular in clinical settings where interpretability is compulsory. Nevertheless, their performance can be limited when the relationships among the variables are weak. Clinical variables can have nonlinear relationships or large input feature spaces, and deep learning models can address these limitations by learning hierarchical features from the input data without manually creating features. The application of deep learning to the Kaggle Heart Disease dataset has been shown to be effective. In particular, multi-layered neural networks have shown competitive performance compared with shallow classifiers, especially when the input space is enriched with polynomial interaction features [25, 26].

There are a number of benchmark studies that have undertaken systematic head-to-head comparisons. A comparison of logistic regression, k-nearest neighbors, decision trees, and random forest on the UCI Cleveland and Kaggle heart data discovered that the ensemble models performed better and consistently performed better than single classifiers [4]. An extensive test on a variety of algorithms and datasets revealed that no single model prevails in all evaluation metrics, which inspired the necessity of hybrid and stacking strategies [27].

Comparisons of ensemble-learning models and AutoML pipelines have highlighted the importance of model selection and hyperparameter optimization in obtaining stable results [28]. Evaluation of several machine learning algorithms for cardiac risk prediction has shown that gradient-boosting methods such as XGBoost can perform well because they can effectively model feature interactions and handle class imbalance [29].

Principal component analysis is a dimensionality reduction method that has been applied and demonstrated a large effect on improving classification by eliminating irrelevant characteristics prior to classification, and pretrained architectures are used in the context of deep learning [30]. Deep learning findings directly encourage the utilization of autoencoder-derived latent representations in the current study, the encoder, which compresses informative and compact embeddings, which are the inputs to downstream classifiers. Beyond tabular data-based approaches, domain-specific data sources have also been investigated [31] and developed a machine learning model based on ECG signal analysis for early detection of congenital heart disease, highlighting the importance of incorporating multimodal medical data.

Stacking and ensemble learning strategies

Stacking is a meta-learning algorithm in which a higher-level meta-learner combines the predictions of multiple base models. This method has gained popularity in medical prediction because it can combine the advantages of different classifiers while reducing individual model bias and error variance [13]. Hybrid ensemble learning has also been used to predict the risk of coronary heart disease. The use of forward feature selection combined with stacked classifiers showed that stacking outperformed any base model individually by a steady difference [13]. It was demonstrated that in ensemble pipelines using SMOTE-based data augmentation, the use of data augmentation led to improved results. Pre-stack sensitivity is greatly enhanced by dealing with class imbalance before stacking in heart disease prediction [32].

This work is closely related to the approach taken in the current experiment, in which a sparse autoencoder with adaptive feature boosting is used, to produce enriched latent representations, which are used as inputs to stacked base learners. Under the suggested CA-SAE-AFB, logistic regression on an autoencoder. Latent features and XGBoost acting on features of interactions of a polygonal nature work as complementary base models. Their probability outputs are then combined. A Gradient Boosting meta-learner then constructs the stacking pipeline. This design is informed by previous findings that the integration of neural and tree-based learners in a model is encouraged. The stacking framework obtains strong generalization on numerous random splits [13, 28].

Autoencoder-based feature learning

Autoencoders are unsupervised neural network architectures that learn compact latent-space representations by reconstructing input data using a bottleneck layer. In the tabular medical data context, autoencoders are potent feature extractors. Autoencoders can reduce noise, project data into lower dimensions, and reveal latent organization that is not evident in the original feature space [9, 10]. A simple autoencoder-based prediction model of heart diseases showed that sparsity regularization causes the encoder to learn to be more selective and meaningful in activation functions [9]. A subsequent study also used particle swarm optimization to optimize sparse autoencoder parameters while maintaining consistency in the division of datasets [33].

The autoencoder is trained using a combined autoencoder and DenseNet architecture, where the autoencoder learns features before DenseNet and reduces the feature space, while DenseNet performs classification using dense layers. This approach takes advantage of supervised deep classification and representation learning together with unsupervised learning [17]. This line of research was advanced by using autoencoders for feature engineering, where the latent space is regarded as an engineered set of features that are forecasted by machine learning models rather than being a direct input to a classifier [10].

The proposed model is based on these assumptions, but it has an adaptive feature boosting aspect added to it. An additional mechanism is introduced into the autoencoder architecture. In particular, an activated weighting layer with 64 dimensions is applied to the latent space. It adjusts each latent dimension according to its estimated relevance. This produces a representation that prioritizes the most distinguishing characteristics. Channel attention modules in computer vision serve as the foundation for this attention-like function and this idea is adapted to the tabular medical domain. autoencoders that do not explicitly account for latent dimensionality. Conditional modeling has also been explored using conditional variational autoencoders. These models learn feature distributions conditioned on class labels and can generate class-aware representations Create representations that take classes into consideration [34]. While variational methods provide probabilistic representations, the proposed approach uses deterministic sparsity. The adaptive feature boosting version of autoencoders applied in the current work offers an easier and simpler version.

Recent medical-AI studies further motivate the use of adaptive feature optimization, efficient representation distillation, and multi-scale adaptive representation learning for clinical prediction. Adaptive feature optimization has been investigated for diabetes prediction, while difficulty-aware feature distillation has been proposed to improve efficient cardiac MRI classification. Self-supervised multi-scale adaptive convolutional-transformer learning has also been explored for robust breast-cancer histopathology classification [35–37]. These studies support the broader methodological motivation for adaptive representation learning and efficient clinical inference, while the present work focuses specifically on leakage-controlled tabular heart-disease prediction and nested evaluation.

Threshold optimization and decision boundary strategies

Many classification studies report performance using a default probability threshold, commonly 0.5, which implicitly assigns the same decision boundary regardless of the relative costs of false-positive and false-negative errors. This assumption is not necessarily appropriate for medical diagnosis, where missing a true positive can have substantially greater clinical consequences than generating a false positive. Therefore, decision-threshold optimization is an important component of clinically oriented classification [11, 38].

A combination of hyperparameter and decision-threshold optimization can improve threshold-dependent performance, particularly when class distributions are imbalanced and the operating point affects the precision–recall trade-off [38]. Threshold selection based on ROC analysis and Youden's index has also been investigated for coronary artery disease risk assessment [39], while explainable-AI studies have emphasized the importance of transparent decision criteria for clinical trust [11].

The CA-SAE-AFB framework therefore avoids imposing a fixed 0.5 threshold. Instead, the optimal decision boundary is obtained by searching $\mathcal{T} = \{0.25, 0.26, \ldots, 0.70\}$ and selecting the threshold that maximizes the F1-score on the appropriate inner-validation/OOF predictions. The selected threshold is then fixed before evaluation on the untouched outer-test fold. A two-stage decision zone is subsequently used to handle borderline predictions through the specified confidence margin. The 50-seed experiment is retained as a stability analysis across predefined stratified random splits and is not treated as an independent estimate of generalization (Table 1).

Research gaps and limitations

There are various gaps in the research on heart disease prediction. The feature interaction concept of modeling is not exhaustively studied. Although feature engineering and autoencoders enhance representation, explicit modeling of feature expansions based on polynomials is frequently restricted. This can restrict the capability of the model to model very complicated associations amid clinical variables like age, cholesterol, resting blood pressure, and ST depression, which are all present in the Kaggle Heart Disease dataset. The other gap is the absence of strong evaluation methods. Few works consider a series of random divisions to guarantee good and steady results. Single-split evaluations are capable of generating biased performance predictions, which fail to predict hidden information. This is handled through the evaluation of 50 stratified random seeds in the present work.

Hybrid stacking approaches between tree-based ensembles and deep learning encoders. Classifiers are yet to be used. Despite some of the studies being based on hybrid models, there is a sparse autoencoders, XGBoost, and gradient boosting limited work on their combination. meta-learners within a single stacking architecture [13, 34]. The proposed hybrid architecture is designed to address this gap.

Threshold optimization is also not often discussed. Many models use a default classification threshold, which might not be the best to make medical decisions [38, 39]. Threshold optimization can improve classification performance by selecting an operating point based on the F1-score using inner-validation/OOF predictions, as implemented in the current work. Lastly, issues such as model interpretability, computation performance, reliability,

Table 1 Comparison of existing studies and proposed CA-SAE-AFB model

Existing Work	Gap	Key Idea	Methodology
[4]	Limited nonlinear modeling	ML models for prediction	Polynomial interaction features
[5]	Feature interaction ignored	Machine learning models	Enhanced feature engineering
[8]	Manual feature design	Feature engineering	Autoencoder-based learning
[15]	Early NN limitations	Multilayer perceptron	Improved deep feature extraction
[6]	Overfitting in DL models	Neural networks for diagnosis	Sparse autoencoder regularization
[7]	Limited generalization	CNN/LSTM architectures	Hybrid DL + ML framework
[16]	Complex architecture tuning	Deep learning models	Simplified hybrid model
[9]	No feature importance	Sparse autoencoder	Adaptive Feature Boosting (AFB)
[10]	Limited feature usage	Autoencoder feature extraction	Enhanced latent representations
[13]	Weak hybrid stacking	Ensemble learning	AE + LR + XGBoost stacking
[27]	No universal best model	Comparative analysis	Hybrid stacking strategy
[32]	Only imbalance handling	SMOTE + ensemble	Threshold optimization
[28]	No DL + ML integration	AutoML ensembles	Custom stacking pipeline
[38]	Fixed threshold usage	Threshold optimization	F1-based tuning
[39]	Limited decision strategy	ROC-based threshold	Two-stage decision boundary
Proposed Work	Addresses identified gaps	Hybrid deep + ML stacking	CA-SAE-AFB + LR + XGBoost + GB with F1-based threshold

and healthcare system integration are not sufficiently covered in current studies [11]. Addressing these challenges is important for translating predictive models into practical clinical tools.

System model / architecture

Overall system architecture

Figure 2 represents the overall architecture of the proposed framework. The system is designed as a hybrid pipeline, which combines deep feature learning with ensemble-based prediction to enhance the accuracy and reliability of heart disease prediction. Raw clinical data is obtained at the start of the process from a publicly available dataset [40]. This data is subjected to a sequence of preprocessing to improve the quality of data and provide consistency. The raw dataset is first preprocessed to make it suitable for model training. One-hot encoding is used to encode categorical variables into numerical representations. To increase the expressive power of the model, polynomial interaction features are created, which allow the relationships between several clinical attributes to be captured. All the features are then standardized to zero mean and unit variance to enable the training of models to be stable and efficient.

Feature interaction and representation

The proposed CA-SAE-AFB model is designed to learn meaningful feature representations while emphasizing the most relevant attributes for classification. Initially, a sparse autoencoder compresses the input data into a lower-dimensional latent space, capturing essential patterns while reducing noise. To further enhance feature importance, an Adaptive Feature Boosting (AFB) mechanism assigns dynamic weights to each latent dimension, similar to an attention mechanism. This allows the model to focus more on discriminative features.

The feature interaction is important in the modeling of complex dependencies in clinical data. Second-order interaction features are created in this work to explicitly depict the pair relationships among variables. This rich feature space enhances the capacity of the model to learn nonlinear patterns that cannot be learned directly in the original input. Figure 3 presents the internal structure of the proposed CA-SAE-AFB model. This architecture is founded on a sparse autoencoder model with an Adaptive Feature Boosting (AFB) mechanism. The encoder reduces the input data (high-dimensional) to a latent representation (lower-dimensional). L1 regularization is used to impose sparsity in the model, which encourages the model to learn only the most useful features. A dropout layer is used to minimize overfitting and enhance generalization. The latent representation is then optimized through the AFB mechanism, which assigns adaptive weights to each of the latent dimensions. This procedure accentuates informative characteristics and reduces the influence of less relevant features, which essentially serves as an attention mechanism in the latent space. The decoder recreates the input information on the boosted latent representation, while retaining the necessary information. At the same time, a classification layer also uses the representation to make predictions about the target output and thus jointly optimizes the reconstruction and classification goals.

Training and validation strategy

To separate model development from final evaluation, the study uses stratified nested cross-validation. A stratified 5-fold outer loop is used to estimate generalization performance, while each outer training partition is further divided using stratified 3-fold inner cross-validation for model development, threshold selection, and generation of out-of-fold (OOF) predictions. The outer test fold is never used for preprocessing, hyperparameter selection, threshold optimization, representation learning, or meta-learner fitting. The 50-seed stratified evaluation is used as a complementary stability analysis rather than the primary estimate of generalization.

https://doi.org/10.1186/s12872-026-06644-5

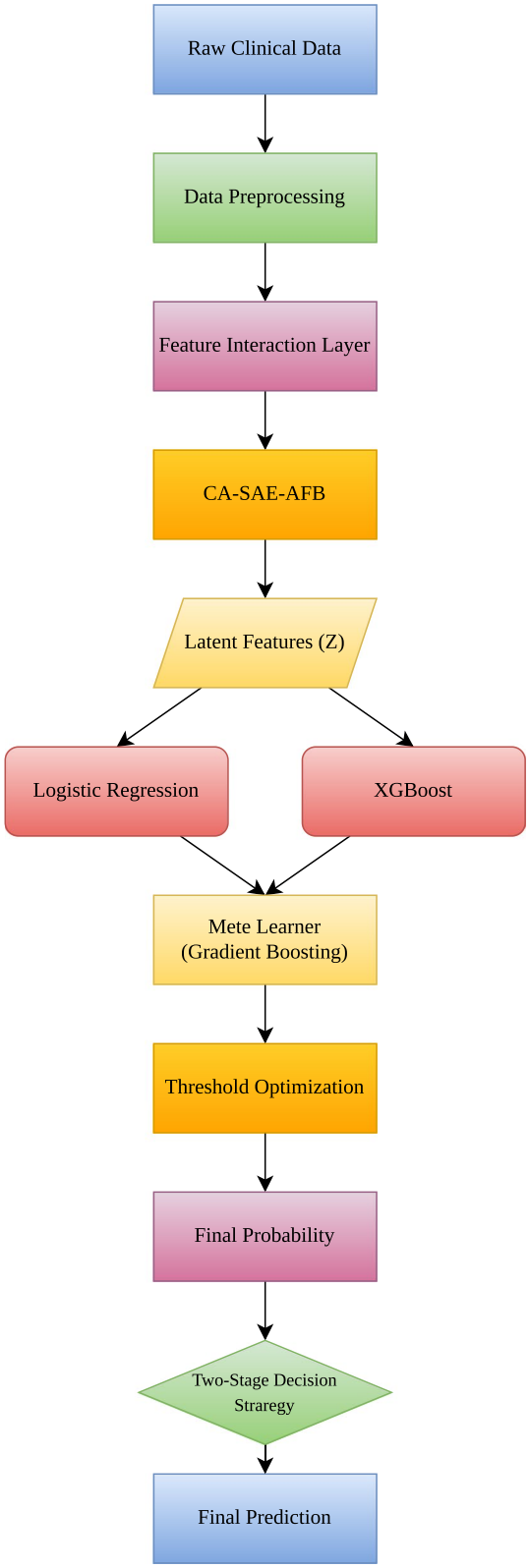


Fig. 2 Overall system architecture of the proposed hybrid prediction framework

All fold-dependent preprocessing operations are fitted using only the relevant training partition. Categorical encoding, interaction-feature generation, and standardization parameters are estimated from the training data and

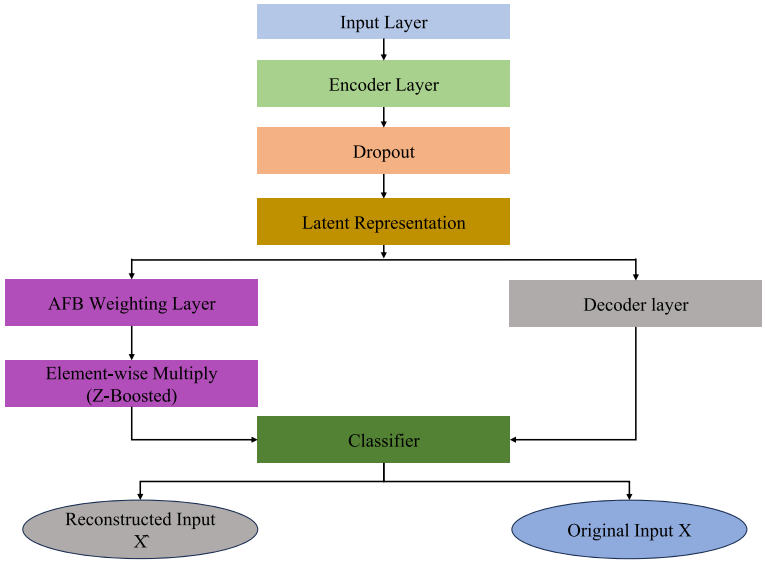


Fig. 3 Internal architecture of the CA-SAE-AFB model showing encoder, AFB weighting, and classification components

then applied unchanged to validation and outer-test observations. No missing observations requiring imputation were identified in the analyzed primary dataset; therefore, no imputation was applied. If missing values are encountered in future datasets, any imputation parameters will likewise be estimated only from the corresponding training partition.

For every inner fold, the sparse autoencoder and Adaptive Feature Boosting (AFB) parameters are learned only from the inner-training observations. The resulting encoder is subsequently used only as a transformation for the corresponding validation observations. After inner-fold development, the outer test observations are transformed using parameters learned from the outer training data and are not used for representation learning.

The stacking ensemble is constructed from OOF base-model probabilities. Logistic regression and XGBoost are fitted on inner-training subsets and generate predictions for observations that were not used to fit the corresponding base learner. These OOF probabilities form the meta-feature matrix for the Gradient Boosting meta-learner. Consequently, no in-sample base-model prediction is used as a meta-learner training target.

The F1-based decision threshold is optimized exclusively on inner validation/OOF predictions according to $t^* = \arg\max_{t \in \mathcal{T}} F_1(y_{\text{val}}, p_{\text{val}}; t)$, where $\mathcal{T} = [0.25, 0.70]$ is searched in increments of 0.01. The selected threshold is fixed before evaluating the untouched outer test fold. Thus, outer-test labels do not participate in threshold selection.

Classification models and ensemble strategy

To effectively leverage both latent and original feature spaces, a hybrid classification strategy is adopted using logistic regression and XGBoost.

Logistic Regression (LR) is employed as a baseline probabilistic classifier operating on the learned latent representations obtained from the CA-SAE-AFB module. It estimates the probability of heart disease occurrence using a sigmoid function:

$$P(y = 1|x) = \frac{1}{1 + e^{-(w^T x + b)}} \tag{1}$$

where x represents the latent feature vector, w denotes the model weights, and b is the bias term. Due to its simplicity and interpretability, logistic regression provides insight into the contribution of learned features toward prediction.

Extreme Gradient Boosting (XGBoost) is utilized to model complex nonlinear relationships directly from the original feature space. XGBoost constructs an ensemble of decision trees in a stage-wise manner, where each tree corrects the errors of previous ones. The prediction is given by:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i), \quad f_k \in \mathcal{F} \tag{2}$$

where f_k represents the k-th decision tree and $\mathcal{F}$ denotes the functional space of regression trees.

Gradient Boosting (GB) is employed as the meta-learner in the proposed stacking ensemble framework to optimally combine predictions from multiple base models. It is an ensemble technique that builds a strong learner by sequentially adding weak learners, typically decision trees, where each subsequent model focuses on minimizing the errors of the previous ones.

The overall prediction in gradient boosting is expressed as an additive model:

$$F(x) = \sum_{m=1}^{M} \gamma_m h_m(x) \tag{3}$$

where $h_m(x)$ represents the m-th weak learner (decision tree), and γ_m denotes the corresponding weight.

At each iteration, the model minimizes a differentiable loss function by fitting a new learner to the negative gradient (residual errors) of the loss with respect to the current model predictions. This iterative refinement enables gradient boosting to be able to capture complex patterns and interactions among model outputs.

Gradient boosting is used as a meta-classifier in this work, with the input features being the derivation of the prediction probabilities of logistic regression and XGBoost. The meta-learner can use the knowledge of how to combine these predictions, thus improving overall predictive performance and generalization. The two models are stacked to produce the outputs. Latent features are used to create probability scores through logistic regression, and original features are used to create probability scores through XGBoost. These scores are used as inputs to a Gradient Boosting meta-learner, which learns to combine them in the best way to make the final prediction. To further improve classification performance, an optimal decision threshold is established by maximizing the F1-score within a specified range. Also, the two-stage decision mechanism is implemented to deal with uncertain predictions, enhancing the reliability in borderline cases.

Performance evaluation

The performance of the proposed model is evaluated using accuracy, ROC-AUC, F1-score, and Matthews Correlation Coefficient (MCC), together with the additional metrics reported for the nested evaluation. These measures provide complementary views of classification performance, particularly in the presence of class imbalance. The repeated-seed experiment is used to assess performance stability with respect to random data partitioning, whereas generalization is primarily assessed using the leakage-controlled nested cross-validation protocol.

Problem formulation

Mathematical model

Let the Kaggle Heart Disease dataset [40] be represented as:

$$D = \{(x_i, y)\}_{i=1}^{N} \tag{4}$$

where $x_i \in \mathbb{R}^d$ denotes the i-th patient's feature vector, $y \in \{0,1\}$ is the respective class label (lack of heart disease or presence of heart disease), and N is the number of samples. The aim of the suggested framework is to acquire a predictive role:

$$f : \mathbb{R}^d \to \{0,1\} \tag{5}$$

that is capable of making the correct diagnosis of the disease of a patient according to clinical characteristics.

The learning algorithm starts with encoding the original input features into a latent representation that is small by using an encoder network:

$$z = f_{\text{enc}}(x_i) \tag{6}$$

where $z \in \mathbb{R}^k$ represents embedding of the data into a lower-dimensional space that preserves the most informative patterns. This change will minimize noise and redundancy and retain vital clinical data.

In order to further discriminate against the latent space, an Adaptive Feature Boosting (AFB) mechanism is used. The weight of importance in this module to each of the latent dimensions is:

$$w = \sigma(W_a z + b_a) \tag{7}$$

$$z' = z \odot w \tag{8}$$

where $w \in \mathbb{R}^k$ denotes learned feature weights, $\sigma(\cdot)$ is the sigmoid activation function, and $\odot$ represents element-wise multiplication. This is the operation that highlights the relevant features and suppresses less informative features and is essentially an attention mechanism in the latent space.

The enhanced latent representation z' is fed through a decoder network to recreate the original input:

$$\hat{x}_i = f_{\text{dec}}(z') \tag{9}$$

This process of reconstruction makes sure that the latent space still has enough information from the input distribution, which enhances generalization and robustness.

At the same time, prediction is performed with the help of a sigmoid classifier using the same boosted representation:

$$\hat{y}_i = \sigma(W_c z' + b_c) \tag{10}$$

This enables the model to jointly optimize feature learning and classification in a unified framework.

Base learners and meta-learning formulation

The framework combines complementary prediction functions. Logistic regression operates on the boosted latent representation z', while XGBoost operates on the interaction-expanded feature space x. Their probabilities are denoted by

$$p_{\text{LR}} = f_{\text{LR}}(z'), \qquad p_{\text{XGB}} = f_{\text{XGB}}(x). \tag{11}$$

For the stacking stage, the meta-feature matrix is constructed exclusively from OOF probabilities:

$$Z_{\text{OOF}} = [p_{\text{LR}}^{\text{OOF}}, p_{\text{XGB}}^{\text{OOF}}]. \tag{12}$$

The Gradient Boosting meta-learner is then fitted as

$$p_{\text{meta}} = G_{\text{GB}}(Z_{\text{OOF}}). \tag{13}$$

where the base-model predictions used to train G_{GB} are generated from folds in which the corresponding observations were not used for base-model fitting. This construction prevents the meta-learner from learning directly from in-sample base-model errors.

The adaptive feature mechanism is a latent-feature re-weighting operation rather than hard feature elimination. For latent representation $z \in \mathbb{R}^k$, the learned weight vector is

$$w = \sigma(W_a z + b_a), \qquad z' = z \odot w. \tag{14}$$

Thus, each latent dimension is continuously emphasized or attenuated according to its learned relevance.

Decision threshold selection is performed only within the inner validation process. The optimal threshold is defined as

$$t^* = \arg\max_{t \in \mathcal{T}} F_1(y_{\text{val}}, p_{\text{meta,val}}; t), \qquad \mathcal{T} = \{0.25, 0.26, \ldots, 0.70\}. \tag{15}$$

The selected threshold is then held fixed for the corresponding outer-test evaluation.

Constraints and parameter settings

All model-development decisions are constrained to the inner cross-validation stage. The outer test fold is reserved exclusively for final evaluation. The latent dimension is 64, the encoder contains 128 hidden units followed by a 64-dimensional latent layer, dropout is 0.3, and L1 regularization is used to encourage sparse representations. The Adam optimizer uses a learning rate of 0.0005 with a batch size of 32 and a maximum of 200 epochs with early stopping (patience = 10).

Optimization objective

The proposed framework is trained with the help of a multi-task learning goal that simultaneously optimizes the performance of both feature reconstruction and classification. The total loss equation is considered to be:

$$L = \lambda_1 L_{\text{rec}} + \lambda_2 L_{\text{cls}} \tag{16}$$

where λ_1 and λ_2 are weighting coefficients that balance the contribution of reconstruction and classification losses. In this work, $\lambda_1 = 0.5$ and $\lambda_2 = 1.0$, giving higher importance to classification accuracy.

Reconstruction Loss: The reconstruction element makes sure that the latent representation does not lose crucial information from the input data. It is characterized by the mean squared error (MSE):

$$L_{\text{rec}} = \frac{1}{N} \sum_{i=1}^{N} \|x_i - \hat{x}_i\|^2 \tag{17}$$

This term encourages the encoder–decoder network to learn meaningful and compact feature representations.

Classification Loss: The alternative classification element is designed based on binary cross-entropy (BCE), which quantifies the difference between the actual labels and the estimated probabilities:

$$L_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \tag{18}$$

This term guides the model toward accurate prediction of heart disease outcomes.

Optimization Process: The overall training objective is to minimize the combined loss:

$$\min L$$

where L represents total loss. The model is optimized using the Adam optimizer with a learning rate of 0.0005.

Threshold Optimization and Decision Strategy: The model output probabilities are refined using threshold optimization after training. The optimal decision boundary is calculated by optimizing the F1-score, as opposed to using a fixed threshold:

$$t^* = \arg \max_{t \in [0.25, 0.70]} F1(t) \tag{19}$$

In order to enhance the reliability of borderline cases, a two-stage decision mechanism is adopted using a confidence margin $\delta = 0.05$ around the optimized threshold t^*. Predictions with $p_{\text{meta}} < t^* - \delta$ are assigned to the negative class, whereas predictions with $p_{\text{meta}} > t^* + \delta$ are assigned to the positive class. Predictions satisfying $|p_{\text{meta}} - t^*| \leq \delta$ are treated as borderline (uncertain) cases and passed to the second-stage decision rule. Thus, the confidence margin explicitly defines the borderline decision zone without using the outer-test labels for threshold selection.

Proposed methodology

Overview of the proposed approach

The proposed pipeline begins with categorical encoding, fold-local feature transformation, and standardization of the clinical variables. Degree-2 interaction features are generated using parameters fitted on the training partition. The resulting representation is processed by the sparse autoencoder and Adaptive Feature Boosting mechanism to obtain a compact, adaptively re-weighted latent representation. Logistic regression uses this latent representation, whereas XGBoost uses the interaction-expanded feature space. Their OOF probabilities are subsequently supplied to the Gradient Boosting meta-learner. The final probability is converted to a binary decision using a threshold selected exclusively from inner validation predictions, followed by the two-stage decision rule for borderline cases.

The evaluation procedure is organized so that no information from the outer test fold enters preprocessing, latent representation learning, base-model fitting, meta-learner training, or threshold selection. This leakage-controlled workflow is the basis for the primary nested-CV results reported below. Algorithm 1 summarizes the proposed CA-SAE-AFB feature-learning procedure, including latent representation learning, adaptive feature re-weighting, reconstruction, and classification loss optimization.

Require: Training data $X \in \mathbb{R}^{N \times d}$ and labels y
Ensure: Boosted latent representation Z
for each epoch $e = 1$ to E **do**
  for each mini-batch (X_b, y_b) **do**
    $z \leftarrow f_{\text{enc}}(X_b)$
    $w \leftarrow \sigma(W_a z + b_a)$
    $z' \leftarrow z \odot w$
    $\hat{X} \leftarrow f_{\text{dec}}(z')$
    $\hat{y} \leftarrow \sigma(W_c z' + b_c)$
    $L_{\text{rec}} \leftarrow |X_b|^{-1} \sum \|X_b - \hat{X}\|^2$
    $L_{\text{cls}} \leftarrow -|X_b|^{-1} \sum [y_b \log(\hat{y}) + (1 - y_b) \log(1 - \hat{y})]$
    $L \leftarrow \lambda_1 L_{\text{rec}} + \lambda_2 L_{\text{cls}}$
    Update parameters using Adam and back-propagation.
  end for
end for
return $Z \leftarrow z'$

Algorithm 1 CA-SAE-AFB Feature Learning

Require: Dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$
Ensure: Outer-fold predictions and evaluation metrics
for each outer fold $j = 1, \ldots, 5$ **do**
  Split D into outer-training D_j^{tr} and untouched outer-test D_j^{te} using stratification.
  Fit preprocessing transformations using D_j^{tr} only.
  for each inner fold $r = 1, \ldots, 3$ **do**
    Fit the SAE-AFB representation learner using inner-training data only.
    Fit LR and XGBoost on the corresponding inner-training data.
    Generate validation predictions for observations not used in base-model fitting.
    Store OOF probabilities p_{LR}^{OOF} and p_{XGB}^{OOF}.
  end for
  Train the Gradient Boosting meta-learner using only $Z_{OOF} = [p_{\mathrm{LR}}^{OOF}, p_{\mathrm{XGB}}^{OOF}]$.
  Select t^* by maximizing F1 on inner validation/OOF predictions.
  Refit the required model components on D_j^{tr} using the selected inner-CV configuration.
  Transform D_j^{te} using training-fitted preprocessing and representation parameters.
  Predict D_j^{te} using the fixed threshold t^* and the two-stage rule.
  Record Accuracy, Balanced Accuracy, Precision, Recall, F1, MCC, ROC-AUC and PR-AUC.
end for
Separately repeat the original stratified train/test evaluation for 50 predefined seeds as a stability analysis.
return outer-fold metrics and stability results

Algorithm 2 Leakage-Controlled CA-SAE-AFB Nested Evaluation

Workflow and operational phases

Algorithm 2 summarizes the leakage-controlled nested-CV evaluation procedure, including fold-local preprocessing, OOF prediction generation, meta-learner training, validation-based threshold selection, and outer-test evaluation. The operational workflow consists of six controlled stages: (1) fold-local preprocessing and degree-2 interaction generation; (2) sparse latent representation learning; (3) adaptive latent-feature re-weighting; (4) base-model prediction using logistic regression and XGBoost; (5) OOF stacking through the Gradient Boosting meta-learner; and (6) validation-only threshold optimization followed by the two-stage decision rule. The outer test fold is accessed only after all model-development decisions have been finalized.

Implementation details

Simulation environment and tools

The experiments were implemented in Python using NumPy, Pandas, TensorFlow, Scikit-learn, and XGBoost. Model development and the final latency experiment were executed in Google Colab using an NVIDIA Tesla T4 GPU. For the latency experiment, 50 warm-up iterations were discarded before 200 timed single-sample inference runs. GPU synchronization was performed before and after timed inference to reduce asynchronous execution effects. The reported latency therefore represents the complete single-sample inference pipeline rather than only the neural-network forward pass.

Table 2 Mann–Whitney U test results for continuous features

Feature	U Statistic	Adjusted p-value	Cohen's d	95% CI (Group 0 / Group 1)
Oldpeak	55164.0	4.06×10^{-36}	-0.887	[0.342, 0.479] / [1.181, 1.373]
MaxHR	153089.5	4.52×10^{-34}	0.878	[145.88, 150.57] / [125.55, 129.55]
Age	69137.5	3.61×10^{-18}	-0.591	[49.68, 51.46] / [55.19, 56.66]
FastingBS	80466.0	8.66×10^{-16}	-0.557	[0.078, 0.137] / [0.293, 0.376]
Cholesterol	121000.5	2.74×10^{-5}	0.481	[220.03, 233.95] / [164.93, 186.92]
RestingBP	90420.0	5.65×10^{-4}	-0.217	[128.67, 131.78] / [132.52, 135.83]

Table 3 Chi-square test results for categorical features

Feature	χ^2 Statistic	Adjusted p-value	Cramér's V
ST_Slope	355.92	2.58×10^{-77}	0.623
ChestPainType	268.07	2.02×10^{-57}	0.540
Exercise Angina	222.26	4.85×10^{-50}	0.492
Sex	84.15	5.75×10^{-20}	0.303
RestingECG	10.93	4.23×10^{-3}	0.109

Dataset description

The primary dataset is the 918-record Heart Failure Prediction dataset used for the main evaluation. It contains 11 clinical predictors and a binary heart-disease outcome. Categorical variables are converted into numerical representations before feature generation. No missing observations requiring imputation were identified in the analyzed primary dataset; consequently, no imputation was applied. The validation protocol enforces fold-local preprocessing so that any preprocessing parameter required for another dataset is estimated only from the corresponding training partition.

A secondary cardiovascular dataset containing 70,000 records is retained for cross-dataset analysis. The reduced performance observed on this dataset is interpreted as evidence of domain shift rather than as proof of universal cross-domain generalization.

The preprocessing pipeline produced 15 model-ready predictors after categorical encoding and feature preparation. The dataset contained 918 observations, with 55.34% positive and 44.66% negative cases. No missing observations requiring imputation were identified in the primary dataset used in this study. Therefore, no imputation was performed during the reported experiments. A fold-local median-imputation procedure is defined only as a safeguard for future datasets and was not used to generate the reported results.

Statistical analysis

To evaluate the discriminative capability of features and support the design of the proposed framework, a comprehensive statistical analysis was performed. Non-parametric hypothesis testing, effect size estimation, and information-theoretic measures were employed. Specifically, the Mann-Whitney U test was used for continuous variables, while the chi-square test was applied to categorical features. Mutual information (MI) was additionally computed to capture nonlinear dependencies. To control for multiple comparisons, the Benjamini-Hochberg procedure was used for p-value adjustment.

Feature significance analysis

The statistical results for continuous and categorical features are presented in Tables 2 and 3. All continuous features were statistically significant ($p_{adj} < 0.001$). Among them, *Oldpeak* and *MaxHR* exhibit large effect sizes, indicating strong class separation, while *Age* and *FastingBS* show moderate-to-large effects. In contrast, *RestingBP* demonstrates a relatively small effect size.

Table 4 Mutual information scores (Mean ± Std)

Feature	MI Mean & MI Std
ST_Slope	0.205 ± 0.010
ChestPainType	0.153 ± 0.011
Exercise Angina	0.132 ± 0.015
Oldpeak	0.116 ± 0.010
MaxHR	0.092 ± 0.009
Cholesterol	0.070 ± 0.008
Sex	0.048 ± 0.012
Age	0.046 ± 0.011
FastingBS	0.036 ± 0.013
RestingBP	0.017 ± 0.012
RestingECG	0.005 ± 0.007

Table 5 Cross-dataset performance comparison of the proposed model under the original benchmark evaluation

Dataset	Accuracy	ROC-AUC	F1 Score	MCC
Dataset 1 [40] (Primary)	0.9203	0.9157	0.9299	0.8400
Dataset 2 [41] (Secondary)	0.7292	0.7987	0.7143	0.4595

Categorical features such as *ST_Slope*, *ChestPainType*, and *ExerciseAngina* exhibit strong associations with the target variable, while *RestingECG* shows a comparatively weak association.

Mutual information and implications

The mutual information results, shown in Table 4, confirm the presence of nonlinear dependencies between several features and the target variable. Features such as *ST_Slope*, *ChestPainType*, and *ExerciseAngina* exhibit the highest information gain, followed by *Oldpeak* and *MaxHR*.

Overall, the statistical analysis demonstrates that the dataset contains strongly discriminative features with both linear and nonlinear relationships. These findings justify the use of hybrid modeling strategies, including tree-based learning for categorical interactions and representation learning for capturing nonlinear patterns. Furthermore, the presence of features with varying importance supports the use of adaptive feature weighting mechanisms in the proposed framework.

The results presented in Table 5 show a substantial reduction in performance on the secondary dataset. This reduction is consistent with domain shift, differences in feature distributions, and differences in clinical populations. Rather than treating the secondary result as evidence of universal transferability, we interpret it as evidence that the framework retains useful predictive information while remaining sensitive to domain-specific characteristics. External prospective validation on harmonized clinical cohorts is therefore required before claims of cross-domain clinical generalization can be made.

Parameter settings

The primary evaluation uses stratified nested cross-validation with a 5-fold outer loop and a 3-fold inner loop. The original 50-seed stratified split evaluation is retained only as a stability analysis. Within every outer fold, preprocessing, SAE fitting, threshold optimization, and meta-learner fitting are confined to the outer-training data and its inner folds. The outer test fold is never used for model selection (Table 6).

The baseline models were evaluated on the same outer folds and with the same fold-wise preprocessing principle. Their results are therefore directly comparable at the evaluation-protocol level. Exact configurations used for the baseline experiments are documented in the accompanying reproducibility materials.

Table 6 Evaluation and model parameter settings

Parameter	Value / Description
Primary evaluation	Stratified 5-fold outer CV with stratified 3-fold inner CV
Stability analysis	50 predefined random seeds using stratified train/test splits
Feature engineering	Degree-2 interaction features, generated within training folds
Scaling	Standardization fitted on training folds only
Encoder	128 hidden units, 64 latent units
Dropout	0.3
Regularization	L1 sparsity regularization
Optimizer	Adam, learning rate = 0.0005
Batch size	32
Epochs	Maximum 200, early stopping patience = 10
Loss weights	Reconstruction = 0.5; classification = 1.0
LR	max_iter = 3000
XGBoost	1000 trees, depth = 6, learning rate = 0.03
Meta learner	Gradient Boosting, 200 estimators, trained on OOF probabilities
Threshold	Inner-validation F1 maximization over 0.25–0.70
Two-stage margin	0.05

Performance evaluation and results

Performance metrics

The evaluation reports Accuracy, Balanced Accuracy, Precision, Recall, F1-score, Matthews Correlation Coefficient (MCC), ROC-AUC, and PR-AUC. For the nested evaluation, mean and standard deviation are calculated across the five outer folds and 95% confidence intervals are reported using the Student-*t* distribution with four degrees of freedom. Paired t-tests and Wilcoxon signed-rank tests are applied to fold-level paired comparisons where appropriate.

Experimental scenarios

Three complementary scenarios are considered. First, the original 50-seed stratified train/test experiment is retained to quantify sensitivity to random partitioning. Second, stratified 5-fold outer and 3-fold inner nested cross-validation is used as the primary leakage-controlled evaluation. Third, component-wise ablation experiments are repeated under the same outer folds to isolate the contribution of SAE, AFB, interaction features, the base learners, the meta-learner, threshold optimization, and the two-stage decision mechanism.

The distinction between the two evaluation schemes is important: the 50-seed experiment characterizes stability under repeated random partitions, whereas nested cross-validation separates model development from final evaluation and therefore provides the more conservative estimate of generalization.

Results and analysis

Figure 4 provides a comparative analysis of all the baseline models and the suggested CA-SAE-AFB framework on four significant evaluation measures, such as accuracy, ROC-AUC, F1-score, and Matthews Correlation Coefficient (MCC). These metrics capture different aspects of model performance, including its overall accuracy, discrimination ability, precision–recall trade-off, and the ability to adapt to the existence of class imbalance.

The nested-CV evaluation provides a more conservative estimate than the best-seed result from the 50-seed benchmark. Under the nested protocol, CA-SAE-AFB achieved mean accuracy of 0.8617, F1-score of 0.8772, MCC of 0.7208, ROC-AUC of 0.9153, and PR-AUC of 0.9196. Conventional tree-based baselines remain highly competitive, and XGBoost achieved higher mean accuracy, F1-score, and ROC-AUC in the same-fold comparison.

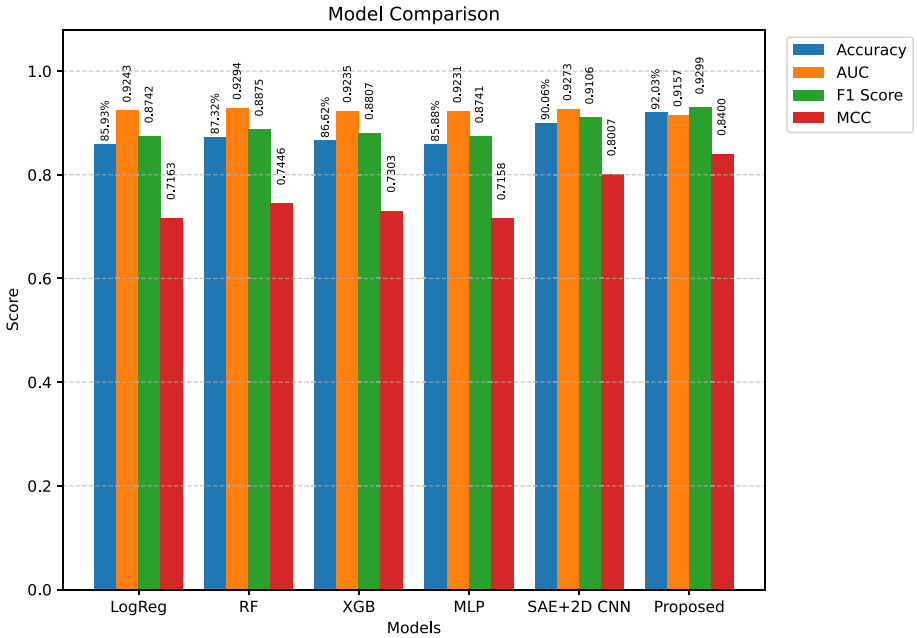


Fig. 4 Performance comparison under the original evaluation; primary conclusions are based on the nested-CV analysis

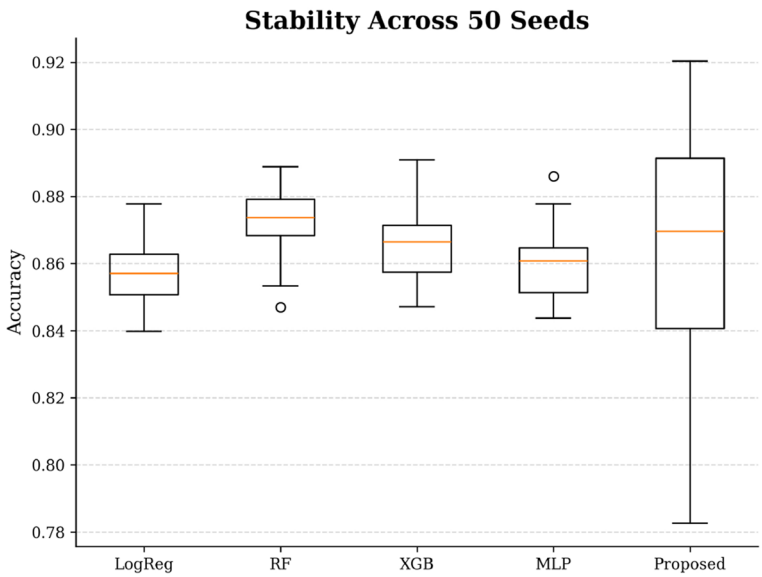


Fig. 5 Stability analysis across 50 random seeds; the nested-CV results are used as the primary generalization estimate

Therefore, the study does not claim universal superiority of CA-SAE-AFB over all baselines. Instead, the contribution is interpreted in terms of a leakage-controlled hybrid architecture and component-level analysis.

To assess sensitivity to random partitioning, the experiment was repeated using 50 predefined random seeds. This analysis is retained as a stability assessment rather than as the primary unbiased estimate. The 50-seed benchmark recorded a best observed run of 92.03% accuracy, 0.9299 F1, 0.8400 MCC, and 0.9157 ROC-AUC; this best-run observation is reported separately from the mean stability analysis. The 50-seed mean estimates for CA-SAE-AFB were Accuracy = 0.86105, F1 = 0.87602, MCC = 0.72010, and ROC-AUC = 0.91162. The corresponding nested 5-fold estimates were 0.86169, 0.87716, 0.72080, and 0.91526, respectively. The small absolute differences indicate close agreement between the repeated-split stability estimate and the stricter nested-CV estimate (Fig. 5).

Further insights into classification performance are provided by the confusion matrix in Fig. 6. The model has 464 true positives and 381 true negatives, with 29 false positives and 44 false negatives. This indicates that the model can correctly classify both negative and positive cases with a relatively low level of misclassification. The balanced classification performance is also consistent with the MCC value reported above.

Overall, the results support methodological robustness and computational feasibility, but they do not establish universal superiority or clinical readiness. The observed domain shift and the absence of prospective clinical validation require the framework to be considered a research-stage decision-support model rather than a clinically deployable system.

Figure 7 illustrates the threshold–F1 relationship from the 50-seed stability analysis and is provided as an illustrative visualization. It is not used to determine the threshold for the primary nested-CV results. In the nested evaluation, the threshold is selected separately within each outer fold using inner validation/OOF predictions. The five selected thresholds were 0.56, 0.46, 0.34, 0.32, and 0.25. The corresponding outer-test observations were evaluated only after the threshold had been fixed.

The Receiver Operating Characteristic (ROC) curves are used to assess the discriminative potential of the models and are shown in Fig. 8. The trade-off between the true positive rate and false positive rate over various thresholds is represented by the ROC curve. Figure 8 provides a representative visualization of the benchmark ROC curves, whereas the 50-seed summary values are reported separately in Table 7. ROC-AUC is not always the most informative measure when class distributions are imbalanced; therefore, the ROC results are considered together with F1-score and MCC.

Further evaluation is performed using precision-recall (PR) curves, as shown in Fig. 9. Because PR curves emphasize the performance of the positive class, they are especially useful in imbalanced datasets. The proposed CA-SAE-AFB model maintains a stable and competitive precision-recall trade-off across different recall values.

Although some baseline models exhibit slightly higher precision at specific recall levels, the proposed model demonstrates smoother and more consistent behavior across the entire curve. This indicates competitive performance in identifying positive cases across different recall levels. The PR curve also provides supporting evidence of the proposed hybrid architecture's performance on the imbalanced clinical data.

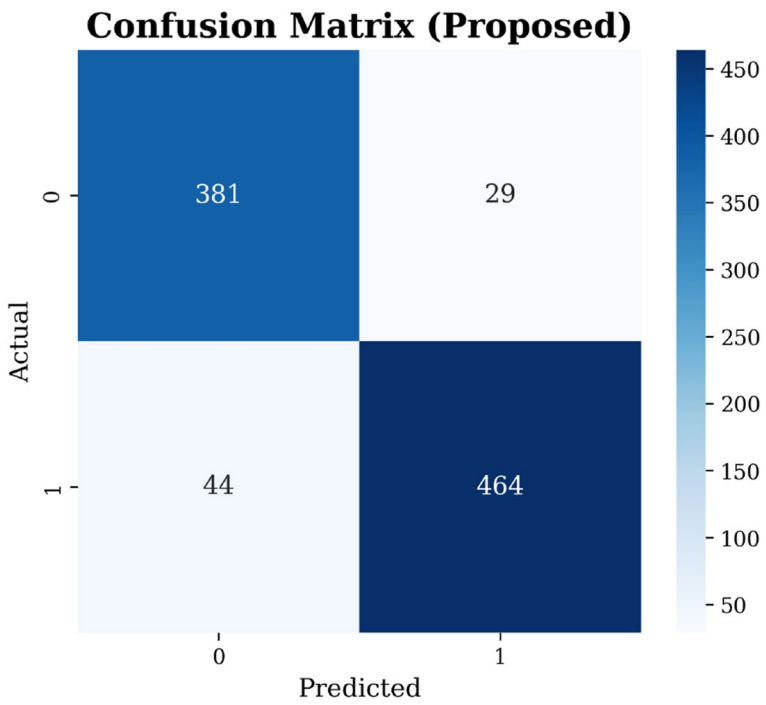


Fig. 6 Confusion matrix of the proposed model from the 50-seed benchmark evaluation

https://doi.org/10.1186/s12872-026-06644-5

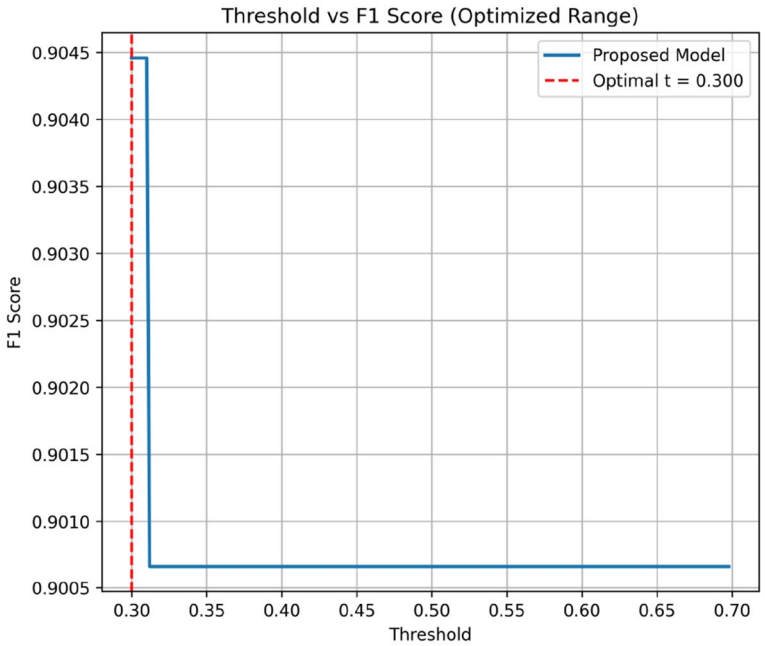


Fig. 7 Illustrative F1-score variation across thresholds from the original stability analysis; nested-CV thresholds are selected independently within inner validation folds

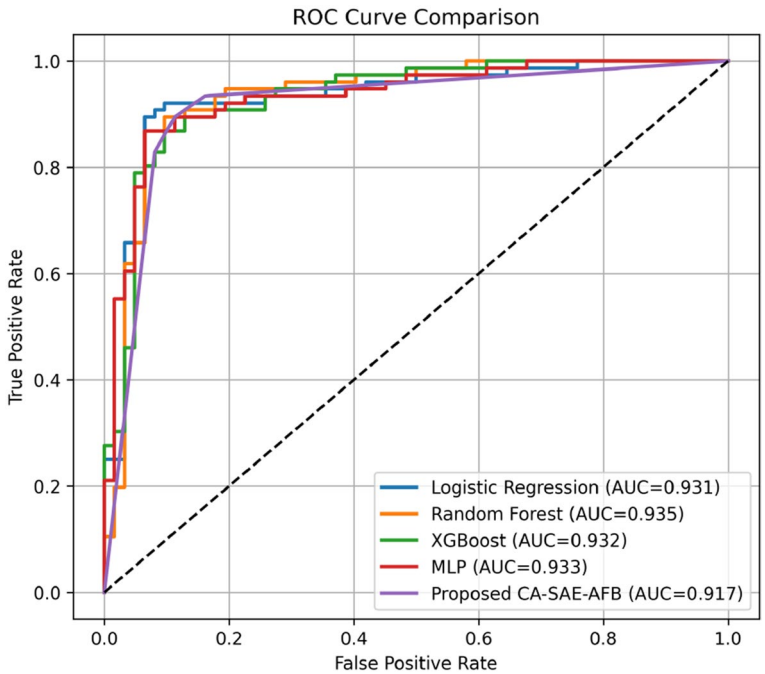


Fig. 8 Representative ROC curve comparison from the benchmark visualization; the 50-seed summary values are reported separately in Table 7

Table 7 Benchmark comparison under the 50-seed evaluation

Method	Accuracy (%)	ROC-AUC	F1 Score	MCC
Logistic Regression	85.93	0.931	0.8742	0.7163
Random Forest	87.32	0.935	0.8875	0.7446
XGBoost (Standalone)	86.62	0.932	0.8807	0.7303
MLP (Deep Neural Network)	85.88	0.933	0.8741	0.7158
SAE+2D CNN	90.06	0.9273	0.9106	0.8007
CA-SAE-AFB	**92.03**	**0.9157**	**0.9299**	**0.8400**

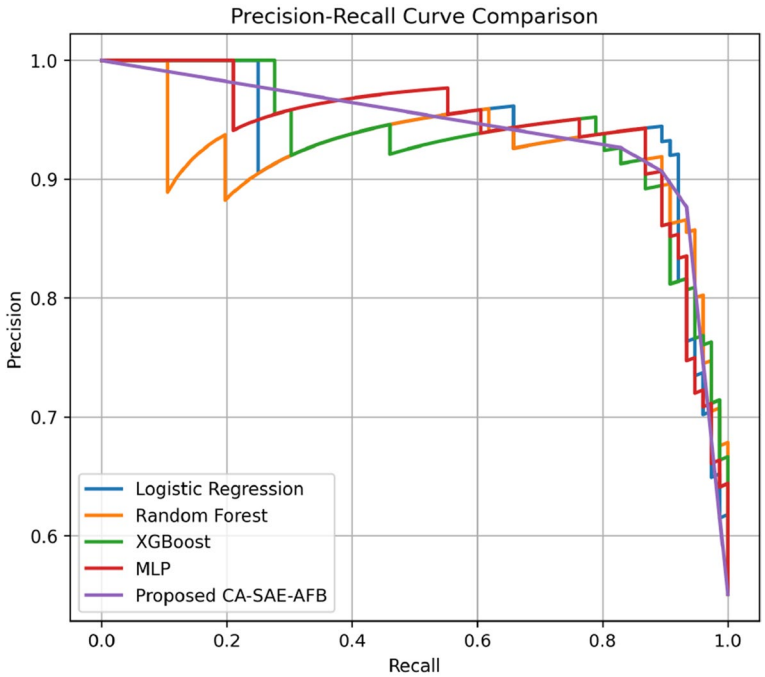


Fig. 9 Precision–Recall curve comparison of baseline models and the proposed method from the benchmark visualization

Table 8 Component ablation under the 50-seed evaluation

Model	Accuracy	AUC	F1	MCC
FULL MODEL	**0.9203**	0.9157	**0.9299**	**0.8400**
NO_INTERACTION	0.8986	0.8945	0.9103	0.7951
NO_SAE	0.8806	0.9321	0.8875	0.7623
NO_AFB	0.8807	0.9318	0.8880	0.7630
NO_META	0.9058	0.9389	0.9193	0.8144
NO_THRESHOLD	0.9203	0.9157	0.9299	0.8400
NO_2STAGE	0.9203	0.9157	0.9299	0.8400

50-seed benchmark evaluation

The 50-seed benchmark provides an experimentally obtained reference for the behavior of the proposed framework under repeated random partitions. It is distinguished from the nested-CV analysis because the two protocols serve different purposes: the 50-seed experiment describes observed benchmark performance and partition stability, whereas nested cross-validation provides the primary leakage-controlled estimate of generalization.

Under the 50-seed benchmark, CA-SAE-AFB achieved 92.03% accuracy, 0.9299 F1-score, and 0.8400 MCC, while Random Forest achieved a slightly higher ROC-AUC (0.9294 versus 0.9157). These values are reported as experimentally observed benchmark results. They are not used to claim universal superiority; the nested-CV estimate provides the primary evaluation of generalization.

Component ablation under the 50-seed evaluation

The component-removal experiment provides complementary evidence under the 50-seed evaluation protocol and is distinguished from the leakage-controlled nested-CV ablation (Table 8).

The 50-seed ablation indicates that interaction features and the meta-learner were associated with reductions in several threshold-dependent metrics when removed, whereas SAE and AFB did not produce a measurable decrease under this original evaluation. The unchanged values for the no-threshold and no-two-stage

Table 9 Asymptotic computational complexity of the CA-SAE-AFB pipeline

Stage	Complexity	Interpretation
Interaction expansion	$O(np^2)$	Degree-2 interaction generation
SAE encoder	$O(n(ph + hz))$	Dense encoder forward computation
AFB	$O(nz^2)$	Adaptive latent re-weighting
Decoder	$O(nzp)$	Reconstruction
XGBoost training	$O(Tnp \log n)$	Approximate tree construction
XGBoost inference	$O(TD)$	Tree traversal
Meta training	$O(Mn \log n)$	Approximate boosting construction
Meta inference	$O(MD)$	Meta-tree traversal

Table 10 End-to-end single-sample inference latency on NVIDIA Tesla T4

Measure	Value
GPU	NVIDIA Tesla T4
Warm-up runs	50
Timed runs	200
Batch size	1
Mean latency (ms/sample)	83.18
Median latency (ms/sample)	74.57
95% CI (ms/sample)	80.15–86.21
P95 latency (ms/sample)	139.87
P99 latency (ms/sample)	160.70
Maximum latency (ms/sample)	178.44

configurations are retained as reported observations. The nested-CV ablation provides a complementary leakage-controlled assessment of component effects.

Explainability analysis

The current study treats explainability as an important deployment consideration but does not claim a completed SHAP analysis unless the corresponding fold-wise attribution results are available. The adaptive feature mechanism provides a model-internal relevance weighting of latent dimensions, while future work will evaluate SHAP or related attribution methods under the same leakage-controlled validation protocol.

Computational feasibility and complexity analysis

The computational cost was analyzed for both training and inference. Let n denote the number of samples, p the interaction-expanded input dimension, $h = 128$ the hidden dimension, $z = 64$ the latent dimension, E the number of training epochs, T the number of XGBoost trees, D the maximum tree depth, and M the number of Gradient Boosting meta-learner trees. Degree-2 interaction generation requires approximately $O(np^2)$ operations. The dense SAE encoder has forward complexity $O(n(ph + hz))$, while AFB has complexity $O(nz^2)$. Decoder reconstruction is approximately $O(nzp)$ and the latent classifier is $O(nz)$. XGBoost training is approximately $O(Tnp \log n)$ under a conventional tree-construction approximation, while its inference cost is $O(TD)$. Gradient Boosting meta-learner training is approximately $O(Mn \log n)$ and inference is $O(MD)$. Therefore, the dominant per-sample inference terms are the feature transformation, SAE/AFB forward pass, and tree traversal. These expressions are asymptotic approximations because the exact complexity depends on the library implementation and hardware (Tables 9 and 10).

Stage-wise profiling showed that the SAE+AFB representation stage was the dominant inference component (mean 73.60 ms), followed by preprocessing (6.98 ms), XGBoost (1.48 ms), the meta-learner (0.79 ms), logistic regression (0.31 ms), and final decision processing (0.008 ms). These measurements demonstrate computational

Table 11 Nested-CV comparison with baseline models

Method	Accuracy	95% CI	F1	95% CI	MCC	ROC-AUC
Logistic Regression	0.8584	[0.8320,0.8848]	0.8766	[0.8562,0.8970]	0.7147	0.9264
Random Forest	0.8661	[0.8345,0.8976]	0.8835	[0.8593,0.9077]	0.7306	0.9266
XGBoost	0.8759	[0.8499,0.9018]	0.8906	[0.8672,0.9139]	0.7492	0.9329
MLP	0.8552	[0.8291,0.8812]	0.8759	[0.8549,0.8969]	0.7090	0.9192
Gradient Boosting	0.8726	[0.8351,0.9101]	0.8879	[0.8586,0.9172]	0.7441	0.9302
Extra Trees	0.8628	[0.8450,0.8805]	0.8791	[0.8645,0.8937]	0.7230	0.9252
SVM-RBF	0.8704	[0.8577,0.8831]	0.8857	[0.8748,0.8966]	0.7380	0.9267
CA-SAE-AFB	0.8617	[0.8365,0.8869]	0.8772	[0.8591,0.8952]	0.7208	0.9153

Table 12 Component-wise ablation under the nested 5-fold evaluation

Configuration	Accuracy	95% CI	F1	95% CI	MCC	ROC-AUC
FULL	0.8617	[0.8365,0.8869]	0.8772	[0.8591,0.8952]	0.7208	0.9153
NO-SAE	0.8639	[0.8515,0.8762]	0.8820	[0.8727,0.8913]	0.7260	0.9170
NO-AFB	0.8508	[0.8123,0.8893]	0.8674	[0.8337,0.9012]	0.6984	0.9123
NO-INTERACTION	0.8530	[0.8186,0.8874]	0.8730	[0.8475,0.8985]	0.7045	0.9195
NO-LR	0.8606	[0.8347,0.8865]	0.8795	[0.8573,0.9017]	0.7190	0.9172
NO-META	0.8584	[0.8360,0.8809]	0.8755	[0.8556,0.8953]	0.7139	0.9275
NO-THRESHOLD	0.8573	[0.8205,0.8942]	0.8730	[0.8423,0.9037]	0.7112	0.9162
NO-2STAGE	0.8552	[0.8228,0.8875]	0.8728	[0.8451,0.9006]	0.7077	0.9159

feasibility in a GPU-assisted research environment; they should not be interpreted as evidence of clinical deployment readiness.

Discussion

The 92.03% accuracy observed in the 50-seed benchmark is reported as an experimentally observed benchmark value. It should not, however, be interpreted as the primary unbiased estimate of generalization. The nested 5-fold evaluation provides the more conservative estimate because model-development decisions are separated from the outer test folds (Table 11).

The same-fold comparison shows that CA-SAE-AFB remains competitive but does not outperform every baseline. XGBoost achieved the highest mean accuracy, F1-score, MCC, and ROC-AUC among the listed baselines. This finding is explicitly acknowledged rather than interpreted as evidence against the usefulness of the proposed architecture. Instead, the value of CA-SAE-AFB is discussed in terms of its integrated representation-learning, adaptive re-weighting, OOF stacking, and decision-threshold framework (Table 12).

Ablation study

The ablation results show that AFB has the clearest directional contribution: removing AFB reduced accuracy by 0.0109, F1 by 0.0097, and MCC by 0.0224. Removing interaction features reduced accuracy by 0.0087 and MCC by 0.0163, while removing the two-stage mechanism reduced accuracy by 0.0065 and MCC by 0.0131. In contrast, removing the SAE slightly increased several metrics, indicating that its predictive contribution is not statistically demonstrable on this dataset. Similarly, removing the meta-learner increased ROC-AUC and PR-AUC, although the full configuration retained slightly higher threshold-dependent accuracy, F1, and MCC. These findings indicate that component contributions are metric-dependent rather than uniformly additive (Table 13).

Table 13 Paired statistical tests for component ablations

Ablation	Metric	Mean difference	t-test p	Wilcoxon p
NO-AFB	F1	0.0097	0.4437	0.3125
NO-AFB	MCC	0.0224	0.4680	0.4375
NO-INTERACTION	Accuracy	0.0087	0.2707	0.3750
NO-2STAGE	Accuracy	0.0065	0.3057	0.6250
NO-META	ROC-AUC	-0.0123	0.0624	0.1250

Statistical significance analysis

None of the selected component-level differences reached the conventional $p < 0.05$ threshold across the five outer folds. The statistical analysis therefore supports the interpretation of directional component effects without claiming statistically significant superiority of the complete architecture.

Limitations of the study

Several limitations should be considered. First, the primary evaluation is based on a single 918-record clinical dataset, and the secondary dataset demonstrates substantial domain shift. Second, although nested cross-validation reduces model-selection bias, five outer folds provide limited statistical power for paired significance testing. Third, the SAE and meta-learner contributions are metric-dependent and were not statistically significant in the present ablation analysis. Fourth, the study does not implement an additional meta-heuristic hyperparameter optimizer; the reported configurations are predefined and selected within the controlled validation framework. Automated Bayesian or meta-heuristic optimization could be investigated in future work, provided that optimization remains strictly confined to inner validation folds. Fifth, the T4 latency experiment demonstrates computational feasibility in a GPU-assisted research environment but does not establish clinical workflow suitability, prospective safety, calibration, or regulatory readiness. Finally, external prospective validation across heterogeneous clinical populations is required before clinical deployment claims can be made.

Conclusion and future work

This study presents CA-SAE-AFB, a hybrid clinical prediction framework that integrates sparse representation learning, adaptive latent-feature re-weighting, complementary linear and nonlinear base learners, OOF stacking, validation-only threshold optimization, and a two-stage decision mechanism. The nested 5-fold evaluation provides a leakage-controlled estimate of 0.8617 accuracy, 0.8772 F1-score, 0.7208 MCC, and 0.9153 ROC-AUC, with 95% confidence intervals reported for the principal metrics. The nested-CV results also show close agreement with the 50-seed stability analysis.

The ablation study indicates that AFB, interaction features, threshold optimization, and the two-stage mechanism provide directional improvements in several threshold-dependent metrics, whereas SAE and the meta-learner exhibit metric-dependent effects. Conventional tree-based baselines, particularly XGBoost, remain highly competitive under the same-fold nested evaluation. Accordingly, the proposed architecture is presented as a robust and computationally feasible hybrid framework rather than as a universally superior classifier. The final T4 experiment yielded a mean end-to-end single-sample latency of 83.18 ms, supporting feasibility for further research on clinical decision-support workflows.

Future work will focus on external prospective validation, calibration and clinical utility assessment, heterogeneous electronic health-record integration, automated hyperparameter optimization within nested validation, and deployment studies on clinically representative hardware.

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Data availability The dataset analyzed during this study is publicly available and can be accessed from its original public repository. No new data were generated during this study.

Code availability The source code, implementation workflow, and supporting experimental configuration for the CA-SAE-AFB framework are publicly available at https://github.com/krishnasivaprasadm-jpg/cardiovascular_CA-SAE-AFB-a. The repository is provided to support reproducibility of the reported experiments and facilitate independent verification of the implementation.

Declarations

Ethics approval and consent to participate Not applicable. This study uses publicly available, anonymized datasets and does not involve human participants, human tissue, or identifiable personal information. Therefore, ethics approval and consent to participate are not applicable.

Consent for publication Not applicable.

Competing interests The authors declare that they have no competing interests.

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